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Strength Training Improves Body Image and Physical Activity Behaviors Among Midlife and Older Rural Women

Rebecca A. Seguin,

Assistant Professor, Nutritional Sciences, Cornell University, Division of Nutritional Sciences, Ithaca, New York

Galen Eldridge,

Research Associate, Montana State University, Department of Health and Human Development, Bozeman, Montana

Wesley Lynch, and

Professor of Psychology, Montana State University, Department of Psychology, Bozeman, Montana

Lynn C. Paul

Professor, Extension Food and Nutrition Specialist, Montana State University, MSU Extension Service, Bozeman, Montana

Rebecca A. Seguin: rebeccaseguin@montana.edu; Galen Eldridge: galen.eldridge@montana.edu; Wesley Lynch: wlynch@montana.edu; Lynn C. Paul: lpaul@montana.edu

Abstract

The effect of strength training on body image is understudied. The Strong Women Program, a 10-week, twice weekly strength-training program, was provided by Extension agents to 341 older rural women (62±12 years); changes in body image and other psychosocial variables were evaluated. Paired-sample t-test analyses were conducted to assess mean differences pre- to post-program. Strength training was associated with significant improvements in several dimensions of body image, health-related quality of life, and physical activity behaviors, satisfaction, and comfort among rural aging women—an often underserved population that stands to benefit considerably from similar programs.

Purpose

Physical activity is essential to maintaining health, wellness, and physical function, particularly as people age, yet individuals over the age of 50 are more sedentary than any other segment of the adult population (Haskell et al., 2007; He & Baker, 2004; Nelson et al., 2007; von Bonsdorff & Rantanen, 2011). Strength training participation, in particular, is an important element of physical activity for midlife and older women because it helps women maintain muscle and bone mass and reduces risk for numerous chronic diseases.

Research indicates that exercise confers benefits to many dimensions of health, including physical, emotional, intellectual, and social wellness, such as improved sleep, quality of life, and mental health (Penedo & Dahn, 2005; Singh, Clements, & Fiatarone, 1997a, 1997b; Stephenson, 2012). However, much of the exercise research has focused on aerobic activity rather than strength training, and few studies have specifically focused on body image among aging women.

Poor body image is commonly reported among younger women, but research suggests that body image perceptions may also be low in older women (Peat, Peyerl, & Muehlenkamp, 2008). Poor body image and body dissatisfaction are of concern in older women as evidenced by their association with disordered eating, dieting, smoking, and cosmetic surgery (Grogan, Fry, Gough, & Conner, 2009; Haas, Champion, & Secor, 2008; Peat et al., 2008). Dieting and restrained eating may also lead to slower cognitive performance, which is of particular concern in older populations (Green & Rogers, 1995). A more positive body image is associated with better self-esteem, quality of life, physical and emotional well-being, social competence, and self-care and with decreased rates of depression and anxiety (McLean, Paxton, & Wertheim, 2010; Slevec & Tiggemann, 2011). In addition, negative body image can be a barrier to participation in physically active leisure activities (Allender, Cowburn, & Foster, 2006).

Research also shows that social support and access to locations and opportunities, as well as many other factors, are important in engaging rural older adults in physical activity (Kruger, Murray, & Zanjani, 2011; Fitzgerald & Spaccarotella, 2009; Wilcox, Castro, King, Housemann, & Brownson, 2000). Thus, increasing opportunities for rural aging women to participate in supportive, community-based exercise programs hold substantial potential to affect positive changes in many dimensions of health (physical, mental, and emotional) among this population (Pratt & Bowman, 2008; Kang & Russ, 2009).

The Strong Women Program (SWP) is an evidence-based, community strength training program specifically targeted to women aged 40 and older, with the goals of helping them maintain their strength, function, and independence (Seguin et al., 2008). The SWP focuses on increasing women's access to regular strength training opportunities and increasing their knowledge about the importance of regular strength training (Seguin et al., 2008). SWP has been successfully implemented by local Extension Family and Consumer Sciences agents whose expertise is providing research-based, community wellness programs (Pratt & Bowman, 2008; Young, Weinert, & Spring, 2012).

Prior research with SWP participants has demonstrated significant pre-post changes across several parameters of physical fitness, including strength, agility, aerobic fitness, and flexibility (Seguin, Heidkamp-Young, Kuder, & Nelson, 2012). The present research aimed to evaluate pre-post changes in body image, other psychosocial variables, and physical activity behavior to further understand the range of potential benefits among rural aging women.

Methods

Participants

Participants were 341 female residents (94% Caucasian) from nine Montana counties. Their age range was 23-87 years (mean $\pm$ SD = 62 $\pm$ 12). Four percent of the women were under the age of 40, 37% were 40-59, and 59% were 60 years old or older. Average population density of the nine counties is approximately 13 people per square mile, compared to an average national county density of 86 people per square mile (USDA Economic Research Service, 2009).

The Program

The SWP classes last approximately 1 hour and consist of 5 minutes of warm up (e.g., walking, marching in place), 40 minutes of strength training, 5 minutes of balance training, and 5 minutes of cool down (i.e., stretching and flexibility exercises). It is recommended that SWP sessions ideally last 8 to 12 weeks, with two 60-minute classes per week on nonconsecutive days with encouragement for participants to engage in the activities at home once weekly (Seguin et al., 2008; Seguin et al., 2012). For the study reported here, the SWP was taught by Montana State University Extension Family and Consumer Sciences agents at 14 sites across the nine counties. The length of sessions varied from 4 to 12 weeks, with the majority running 10 weeks. The number of participants per county also varied; some counties offered one session, while others offered multiple sessions. Participants per country ranged from seven (one site, one session) to 94 (several sites, multiple sessions), with an average of 32 participants per country.

Measures

At pre- and post-session individual appointments, participants completed questionnaires designed to collect information on several outcomes of interest: psychosocial factors, including body image, and physical activity behavior. Tufts University, in close collaboration with partnering institutions across the U.S. (in the case of the study reported here, Montana State University), conducts ongoing research related to the StrongWomen Program. All materials and procedures for the research were approved by the Tufts University Human Investigation Review Board (#7049).

The *Multidimensional Body-Self Relations Questionnaire (MBSRQ)* assesses multiple attitudinal aspects of body image (Andresen, Fouts, Romeis, & Brownson, 1999; Brown, Cash, & Mikulka, 1990). This validated self-evaluation measure was designed to assess cognitive, behavioral, perceptual, affective, and subjective views about one's body (Thompson, Heinberg, Altabe, & Tantleff-Dunn, 1999). The six subscales (36 total items) used included the following:

- Fitness Evaluation (3 items)
- Fitness Orientation (13 items)
- Health/Illness Evaluation (6 items)
- Health/Illness Orientation (8 items)

- Weight Preoccupation (4 items)
- Self-Classified Weight (2 items)

For Fitness and Health/Illness statements, participants respond using a 5-point scale ranging from 1 (definitely disagree) to 5 (definitely agree). For Self-Classified Weight statements, respondents choose from 1=very underweight, 2=somewhat underweight, 3=normal weight, 4=somewhat overweight, 5=very overweight. The Weight Preoccupation question asks about frequency of crash diets, with the following answer options 1=never, 2=rarely, 3=sometimes, 4=often, 5=very often.

Health-related quality of life was measured using the Centers for Disease Control and Prevention (CDC)'s four-item set of HRQOL questions (Centers for Disease Control and Prevention, 2000). The four HRQOL questions are: (1) "Would you say that in general your health is excellent, very good, good, fair, or poor?"; (2) "Now thinking about your physical health, which includes physical illness and injury, for how many days during the past 30 days was your physical health not good?" (physically unhealthy days); (3) "Now thinking about your mental health, which includes stress, depression, and problems with emotions, for how many days during the past 30 days was your mental health not good?" (mentally unhealthy days); and (4) "During the past 30 days, for about how many days did poor physical or mental health keep you from doing your usual activities, such as self-care, work, or recreation?" (activity-limiting days). The "unhealthy days" measures have been validated in several population studies (Andresen et al., 1999; Moriarty, Zack, & Kobau, 2003; Ounpuu, Krueger, Vermeulen, & Chambers, 2000). They have also been used by the CDC in the Behavioral Risk Factor Surveillance System (BRFSS) surveys since 1993 and in the National Health and Nutrition Examination Survey since 2000 (Moriarty et al., 2003).

Additionally, participants answered questions about comfort level with exercise (2 items). Specifically, they are asked (1) if they are comfortable with their current level of physical activity and (2) if they are comfortable increasing their level of physical activity. The comfort scale, developed by Cardello and colleagues, ranges from -100 (greatest imaginable discomfort) to +100 (greatest imaginable comfort) (Cardello, Winterhalter, & Schutz, 2003).

The Physical Activity Questionnaire, previously used in the Steps to a New You program was used to assess physical activity behavior and satisfaction (Pelican, Paul, Raidl, Gossett, & Lanting, 2006). The first three questions ask about times per week participants engaged in moderate physical activity, stretching, and strength-training, with seven possible answers ranging from 0 to 7 times per week. The two statements "I enjoy being physically active" and "I get as much physical activity as I need" are scored on a five-point scale ranging from 1 (strongly disagree) to 5 (strongly agree). Additionally, there are two questions related to sedentary behavior; these ask about the number of hours per weekend day and per weekday of television viewing. The 5-point scale for these questions ranged from 1 (8 hours or more) to 5 (1 hour or less).

Exercise logs were used by women during each session to keep track of the amount lifted for the six exercises for which hand weights (dumbbells) or adjustable ankle weights were used. Progression is an important component of a strength training program. Participants in the

SWP are encouraged to slowly and safely increase the intensity of the exercises (or the amount of weight they are lifting) to help them continue to improve strength and related benefits.

Analysis

Individual pre- and post-program evaluation scores were used to calculate pairwise t-tests for each dependent variable (DV). The pre- and post-program group means, mean change, 95% confidence interval and p-value associated with each DV are shown in Tables 1-3, where p-values < 0.05 are considered significant.

Results

Average percent attendance of the classes was 74 ± 19 . Although 341 women participated, not every participant completed pre- and post-program questionnaires and/or exercise logs. Overall, the women progressed in terms of intensity in the exercise program, and they improved across several behavioral and psychosocial outcomes following participation in the 10-week community-based exercise program.

Participants significantly increased the weight lifted for all exercises in which hand weights or adjustable ankle weights are used. The complete list of exercises, the order in which they were completed, and the participants' pre-post weights lifted are shown in Table 1.

Six categories from the MBSRQ were assessed in the study, as shown in Table 2. At the conclusion of the program, the women had significantly improved on body image scores for Health Orientation, Subjective Weight, Fitness Orientation, Fitness Evaluation, and Health Evaluation. There was no change in Weight Preoccupation. Participants also reported a small but significant improvement in the General Health measure within the HRQOL (pre-test score=3.45; post-test score=3.55; $p=0.02$); there were no changes in the other measures of the HRQOL (data not shown).

Participants also reported engaging in significantly more frequent moderate physical activity, stretching, and strength training compared to when they started the program. They reported significantly higher levels of physical activity satisfaction—feeling increased levels of enjoyment for physical activity and feeling that they are getting enough physical activity (Table 3). Shown in Figure 1, there was a statistically significant positive increase in comfort with current level of exercise pre- to post-test (item 1). The magnitude of the change was about 30 scale points. In the case of comfort with increasing exercise level (item 2), on the other hand, there was a small but statistically significant decrease in comfort from pre- to post-test. In this case, the decrease was from a relatively high level of positive comfort (59.37) to a slightly lower, but still positive, level (55.01). Despite the fact that on average item 1 tended to increase while item 2 tended to decrease (from pre- to post-test), these scores were strongly positively correlated ($r=0.284$, $p<.001$).

Discussion

Prior research has demonstrated fitness benefits of this community-based strength training program targeted to midlife and older women (Seguin et al., 2012). The present study is the first to demonstrate improvements in body image, health-related quality of life, personal satisfaction, and comfort about physical activity levels as well as physical activity behaviors.

Both experimental and observational studies indicate that exercise has a positive impact on dimensions of body image, including perceived physical appearance, body self-concept, and social physique anxiety (Campbell & Hausenblas, 2009; Hausenblas, Janelle, Gardner, & Hagan, 2003; Tiggemann, 2004). A 2007 meta-analysis by Reel and colleagues also found that weight training more positively affected body concerns than aerobic-type exercises (Reel et al., 2007). Although prior studies examining the effects of strength training on body image have demonstrated improvements, the majority of these studies have been conducted with pre-adolescents, adolescents, or young adults (Ahmed, Hilton, & Pituch, 2002; Caruso & Gill, 1992; Duncan, Al-Nakeeb, & Nevill, 2009; Lubans & Cliff, 2011; Lubans, Sheaman, & Callister, 2010; Martin Ginis, Eng, Arbour, Hartman, & Phillips, 2005).

Only three studies were identified that have examined the effects of strength training programs on body image in middle aged and older adults, with mixed results. Tucker and colleagues found that middle-aged women who participated in a resistance training program improved their body image perceptions significantly more than those in a walking program (Tucker & Mortell, 1993), and more recently, a study by Speck and colleagues reported that twice-weekly strength training for 1 year resulted in significantly greater improvement in body image in a strength-training versus control group among female breast cancer survivors (Speck et al., 2010). However, Shaw and colleagues found no differences in perception of physical appearance between the exercise and control group participants following nine months of strength training (Shaw, Ebbeck, & Snow, 2000). Collectively, these studies indicate that strength training may improve body image among aging women, but the data are not conclusive, and the number of studies is quite limited. Thus, data from the present study add to this literature, demonstrating that several dimensions of body image are sensitive to change among older rural women, including positive shifts in attitudes and beliefs about their weight, fitness, and health.

We know that small increases in time spent being active cumulatively add up to important health benefits (Haskell et al., 2007). Analysis of data from the 2001 BRFSS survey found that adults who engaged in recommended levels of physical activity were significantly less likely to report 14 or more unhealthy days for each of the unhealthy day measures compared to physically inactive adults. These associations were observed across all age groups, although the associations tended to be stronger in adults aged 45 years or older (Brown et al., 2003; Brown et al., 2004). In the present study, data from three physical activity behaviors—general physical activity, stretching, and strength training—all improved, which is a positive indication of healthy lifestyle progress in this population. Future research may benefit from using objective pre-post measures (e.g., accelerometers) of physical activity behavior change during and following strength training program participation in similar populations.

Another important element of change in the present study is the positive shift towards enjoying exercise more. The women still report only moderate success (post-test: 2.85 out of 5) in terms of getting enough exercise, but it is an improvement in the right direction. This finding is reinforced by participants' reported feelings of comfort about their current physical activity level pre and post program, which increased 30 points (out of 200). Mentally, they are feeling more positive about exercise and more positive about themselves in terms of how much exercise they are participating in. We also know that it can be difficult for people to initiate change and maintain it (Seguin et al., 2008). The retention rate was nearly 80%, and the program clearly provided important knowledge and tools to help the women move towards being more active in general, enjoying exercise more, and feeling more comfortable with physical activity. The high retention rate also lends support to the benefits of social settings, including exercising with others and a positive learning environment created by the program facilitators.

Quality of life during the later years has also been found to be as important as duration of life for many older individuals (Branch, 1985). In the study reported here, women improved significantly only on the "General Health" category of the HRQOL. Although this is a positive improvement, there was no change in the other subcategories (i.e., physically unhealthy days, mentally unhealthy days, and activity-limited days). However, the overall number of unhealthy days among this population at baseline was already quite low (with less than 3 days per month, on average, across the three categories at baseline). Thus, the possibility to observe significant improvement was somewhat limited. It is also possible that the intensity or duration of the program was inadequate to elicit change in the other categories. Further studies should include program sessions lasting longer than ten weeks as well as aim to conduct follow-up with participants to examine the durability of benefits.

It is important to note that the present study was limited by the pre-post design; a randomized controlled intervention trial would be preferred but was not possible. Also, there are more robust measures for assessing behavior (e.g., accelerometry, as previously mentioned) as well as additional psychosocial measures that could have been added. However, considerations related to cost as well as the content and length of the questionnaires were critical components of feasibility in the community study. Despite these limitations, the present study makes important contributions to the literature in the area of positive psychosocial improvements, such as body image perceptions, as well as physical activity behavior change and attitudes among aging women. It also lends further support about the feasibility of community-based exercise program implementation *and* outcome evaluation. Perhaps most important, more than 340 women from nine rural counties in Montana were included in the research—a group of individuals who are inadequately represented in the existing literature.

Access to safe and affordable physical activity opportunities are greatly needed in rural communities, and aging women remain one of the most sedentary segments of society. Thus, targeted, evidence-based community exercise programs, which include key social support elements, are free or very low cost, and are feasibly implemented in rural areas (Seguin et al., 2008) hold great promise for improving multiple dimensions of physical, emotional, and mental health among a population with great need that is commonly

underserved and understudied. The Family and Consumer Sciences program area of Extension is an ideal organization to implement community-based strength-training programs, such as the Strong Women Program.

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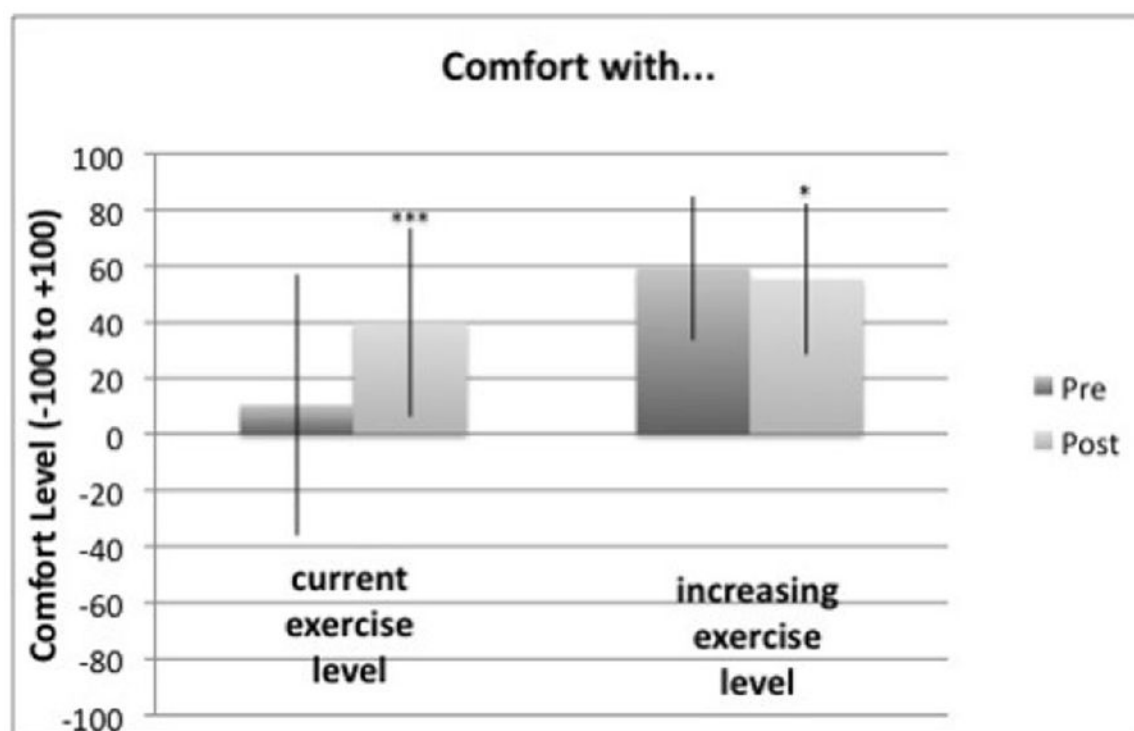


Figure 1.
Change in Participants' Comfort with Exercise Level

Table 1

The Exercise Program and Amount of Weight Being Lifted

Strength Training Exercise	Pre-Test (lbs) (Mean)	Post-Test (lbs) (Mean)	Change (lbs) ($\pm$ SD)	N	95% CI	p-value
Wide leg squat *	-	-	-	-	-	-
Standing leg curl	2.1	4.7	2.6 ($\pm$ 2.4)	218	2.3, 3.0	<0.001
Knee extension	2.1	4.8	2.7 ($\pm$ 2.5)	209	2.4, 3.1	<0.001
Standing side hip/leg raise	2.1	4.6	2.4 ($\pm$ 2.5)	228	2.1, 2.7	<0.001
Biceps curl	2.6	5.4	2.8 ($\pm$ 2.1)	261	2.5, 3.0	<0.001
Overhead press	2.6	4.9	2.3 ($\pm$ 2.0)	252	2.0, 2.5	<0.001
Bent forward fly	2.3	3.9	1.6 ($\pm$ 1.6)	225	1.5, 1.9	<0.001
Toe stand *	-	-	-	-	-	-

* These two exercises use body weight instead of dumbbells or adjustable ankle weights; they are shown here to be inclusive of all exercises that are completed. All exercises in the program are shown above in the order in which they are completed in the program.

Table 2

Multidimensional Body-Self Relations Questionnaire

Body Image Dimension	Pre-Test (Mean)	Post-Test (Mean)	Change (± SD)	N	95% CI	p-value
Health Orientation *	3.69	3.74	0.06 (± 0.45)	254	0.00, 0.11	0.048
Subjective Weight †	3.76	3.64	0.12 (± 0.54)	224	0.05, 0.19	0.001
Fitness Orientation *	3.20	3.44	0.24 (± 0.44)	223	0.19, 0.30	<0.001
Fitness Evaluation *	3.18	3.28	0.10 (± 0.61)	227	0.02, 0.18	0.017
Health Evaluation *	3.65	3.73	0.07 (± 0.43)	226	0.02, 0.13	0.009
Weight Preoccupation ‡	2.75	2.74	0.01 (± 0.55)	225	-0.06, 0.08	0.811

* Higher or increasing score is better for this scale.

† A score of 3 is “normal”; thus, change in the direction of a score of 3 is a positive change.

‡ Lower or decreasing score is better for this scale.

Table 3

Physical Activity Behaviors and Satisfaction

Characteristic	Pre- Test (Mean)	Post- Test (Mean)	Change (± SD)	N	95% CI	P-value
Physical Activity Category (times/week)						
Moderate Physical Activity	3.53	4.26	0.72 (± 1.52)	235	0.53, 0.92	<0.001
Stretching	1.79 1.08	2.93 2.47	1.14 (± 1.73)	235	0.092, 1.36	<0.001
Strength Training	1.08	2.47	1.39 (± 1.57)	235	1.19, 1.60	<0.001
Physical Activity Satisfaction (1=strongly disagree, 5=strongly agree)						
I enjoy being physically active	4.30	4.42	0.113 (± 0.68)	234	0.03, 0.20	.012
I get as much physical activity as needed	2.36	2.85	0.485 (± 0.83)	234	0.38, 0.59	<0.001